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A multisite, randomized, controlled field study of a prepartum cholecalciferol injection on milk production, uterine health, blood minerals, and vitamin D metabolites in multiparous dairy cows

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ABSTRACT

The objective of the present study was to evaluate the effect of a prepartum injection with 10×10^6 IU cholecalciferol on milk production, uterine health, serum total calcium concentration (tCa) and vitamin D metabolites in a multisite setting, on farms feeding a prefresh diet with a positive DCAD. The study was conducted on 11 farms in northwestern Germany. Cows at the end of their first or greater lactation (n = 446) were randomly assigned to a treatment or a control group. Seven days before calving, cows of the treatment group received 10×10^6 IU (10 mL) of cholecalciferol, whereas cows of the control group received 10 mL of saline, intramuscularly. In a subsample of 95 cows, the concentrations of tCa, inorganic phosphate (P_i) and Mg were measured at 1, 2, and 3 DIM. In 30 of these cows, cholecalciferol, 25-OHD_3, 1,25-dihydroxycholecalciferol (1,25-$(OH)_2D_3$), and 24,25-dihydroxycholecalciferol (24,25-$(OH)_2D_3$) concentrations were determined 2 d after injection and on 1, 2, and 3 DIM. At calving, cows were monitored for calving ease and stillbirth, and cows were examined for early postpartum diseases (retained fetal membranes [RFM], clinical hypocalcemia, metritis) until 10 DIM. Repeated measures ANOVA was performed to evaluate the effect of treatment on milk yield over the first 5 test days after calving, blood minerals, and vitamin D metabolites. A logistic regression model was calculated to evaluate the effect of treatment on RFM and metritis. Evaluating milk production over the first 5 test days, there was no difference in milk yield between cholecalciferol treated and control cows. No difference was observed between treated and control cows with regard to gestation length, calving ease, clinical hypocalcemia, and metritis. However, cows treated with cholecalciferol had a greater risk of incurring RFM after treatment with cholecalciferol (odds ratio = 2.12). Cows treated with cholecalciferol had slightly higher serum tCa concentrations with a maximum difference of 0.1 mmol/L at 1 DIM. Furthermore, cows in the treatment group had elevated serum P_i and decreased Mg concentrations over the first 3 DIM. The concentrations of cholecalciferol, 25-OHD_3, 24,25-$(OH)_2D_3$ were increased already 2 d after cholecalciferol treatment and remained higher until 3 DIM, compared with control cows. Although the cholecalciferol concentration decreased considerably after calving, the concentrations of 25-OHD_3 and 24,25-$(OH)_2D_3$ remained around 200 ng/mL, and 20 ng/mL over the first 3 d after calving, respectively. In contrast, the concentration of 1,25-$(OH)_2D_3$ was elevated 2 d after treatment, but decreased after calving, compared with control cows. Overall, in cows fed with a positive DCAD ration prepartum, there were only minor effects of treatment on blood tCa status.
Key words: vitamin D, hypocalcemia, dairy cow

INTRODUCTION

Hypocalcemia around calving is a common metabolic disorder that predisposes the cow to numerous secondary sequela (Couto Serrenho et al., 2021). In the 1970s and 1980s, studies were conducted to evaluate whether cholecalciferol might be injected before calving to increase vitamin D concentrations around calving and prevent cows from clinical hypocalcemia (Gürtler et al., 1977; Littledike and Horst, 1982; Zepperitz et al., 1991). Following these studies, injectable cholecalciferol was approved for milk fever prevention and is commonly

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used in European and Asian countries (Yamagishi et al., 2000; Constable et al., 2017; Venjakob et al., 2017). Cholecalciferol is a fat-soluble vitamin obtained from dietary sources or UV-dependent cutaneous synthesis that is also regarded as a prohormone, as its active form, 1,25-dihydroxycholecalciferol (**1,25-$(OH)_2D_3$**), plays a major role in maintaining Ca and P homeostasis and affects bone metabolism and immune functions (Nelson et al., 2012; Colotta et al., 2017). The first hydroxylation to 25-Hydroxycholecalciferol (**25-OHD_3**) mainly takes place in the liver and is mediated by 2 cytochrome P450 enzymes, CYP2R1 and CYP27A1. This step is substrate-dependent rather than regulated. In contrast, the second step, renal hydroxylation to 1,25-$(OH)_2D_3$ via CYP27B1 is generally accepted to be strictly controlled mainly by parathyroid hormone (**PTH**), ionized Ca (**Ca^{2+}**), phosphate (**P_i**), fibroblast growth factor 23 (**FGF23**), and the concentration of 1,25-$(OH)_2D_3$ itself as negative feedback (Van Etten et al., 2008). As a steroid hormone, 1,25-$(OH)_2D_3$ exerts most of its effects via the genomic pathway after forming a ligand-activated transcription factor together with the vitamin D receptor (**VDR**). It has been estimated that approximately 3% of the genome is directly or indirectly regulated by 1,25$(OH)_2D_3$ (Bouillon et al., 2022). In enterocytes, 1,25-$(OH)_2D_3$ promotes transcription and translation of proteins that ensure active Ca uptake from the gut (Horst et al., 2003). Moreover, 1,25-$(OH)_2D_3$ leads to an increased Ca mobilization from bone (Colotta et al., 2017). Incorporating another OH group at position 24, CYP24A1 initiates the inactivation of 25-OHD_3 to 24,25-$(OH)_2D_3$ and 1,25-$(OH)_2D_3$ to 1,24,25-thihydroxycholecalciferol (Jones et al., 2012). Hydroxylation at position 24 is stimulated by 1,25-$(OH)_2D_3$ itself, FGF23, Ca^{2+} and P_i and inhibited by PTH (Jones et al., 2012; Christakos et al., 2019).

Studies investigating the effect of cholecalciferol treatment mainly focused on the effect on vitamin D metabolites, Ca, and P concentration (Hollis et al., 1977; Naito et al., 1987; Zepperitz et al., 1991). Other authors also studied the effect on milk fever incidence (Julien et al., 1977), and vitamin D toxicity (Littledike and Horst, 1982; Zepperitz et al., 1991). However, there is limited information on health and performance in subsequent lactation in cows treated with high doses of cholecalciferol. Recently, we conducted 2 studies analyzing the effect of prepartum cholecalciferol injections on blood minerals, vitamin D metabolites, reproductive diseases, and milk production on a commercial dairy farm feeding a negative DCAD diet (Venjakob et al., 2022, 2024). Despite a pronounced effect on serum Ca concentration after calving, treated cows showed a reduced milk production, a shortened gestation length (**GL**), a greater risk for retained fetal membranes (**RFM**) and metritis, and compromised reproductive performance (Venjakob et al., 2022). In contrast, an increased milk production has been observed in several studies when feeding cows with 3 mg of 25-OHD_3 daily (Martinez et al., 2018b; Poindexter et al., 2023a).

Beyond its role in mineral metabolism, vitamin D signaling has been increasingly recognized as an important modulator of immune function, inflammatory responses, and tissue integrity in dairy cattle (Merriman et al., 2015; Nelson et al., 2016). Vitamin D receptors are expressed in a wide range of immune and reproductive tissues, and variation in vitamin D status has been associated with altered immune function, cytokine production, and inflammatory signaling (Nelson et al., 2010; Merriman et al., 2015; Martinez et al., 2018a). These mechanisms provide a biologically plausible link between prepartum vitamin D exposure and postpartum uterine health, including RFM and metritis (Kimura et al., 2006; Martinez et al., 2012). Because uterine health during early lactation is closely related to subsequent milk yield and lactational performance (Dubuc et al., 2011; Figueiredo et al., 2024) interventions that modify vitamin D signaling may influence production outcomes through pathways that extend beyond calcium homeostasis alone.

Therefore, the objective of this multisite, randomized controlled field trial was to evaluate the effects of a single intramuscular injection of cholecalciferol administered during late gestation on milk production and postpartum health outcomes in multiparous dairy cows. Milk yield was defined as primary outcome, whereas uterine health, blood minerals, and vitamin D metabolites were considered secondary outcomes. We hypothesized that a single injection in a multisite setting would not reproduce the previously reported negative effects on milk production and uterine health.

MATERIALS AND METHODS

This randomized controlled trial was conducted on 12 commercial dairy farms located in Northwest Germany (federal state of Lower Saxony) between April 2022 and May 2023. The experimental procedures reported herein were approved by the federal authorities of Lower Saxony (protocol 33.19-42502-04-21/3828).

To evaluate the effect of the treatment on milk production (primary outcome), milk production was recorded over the first 5 monthly test days. To evaluate the effect on secondary outcomes, cows were examined for RFM and metritis during the first 10 DIM. The effect on blood minerals was evaluated in a subsample of cows sampled on 1, 2, and 3 DIM. In a further subsample of cows, the effect of treatment on vitamin D metabolites was evaluated. Those cows were sampled at the same time points as described above and a further blood sample was drawn 2 d after treatment application.

Table 1. Chemical composition of the prefresh diets of the 11 farms enrolled in this study (%, unless otherwise noted)

Nutrient composition (DM basis)	Farm 1	2	3	4	5	6	7	8	9	10	11
CP	12.7	13.8	14.8	15.7	15.3	13.8	11.7	16.3	15.5	13.0	13.2
Ether extract	2.6	3.4	3.5	3.9	3.5	3.3	2.5	4.8	3.3	2.7	2.9
NDF	49.4	43.7	46.7	37.6	46.0	42.6	49.6	54.1	49.5	44.4	50.1
NFC[1]	27.7	31.6	23.2	33.6	26.1	32.9	28.8	14.6	15.4	32.6	23.6
Starch	7.6	18.6	6.6	12.9	11.5	22.9	17.4	n.d.	5.5	18.1	10.0
Ash	7.6	7.5	11.8	9.2	9.1	7.4	7.4	10.2	16.3	7.3	10.2
Ca	0.4	0.72	0.64	0.65	0.56	0.85	0.62	0.53	0.74	0.4	0.56
P	0.27	0.32	0.32	0.33	0.34	0.36	0.39	0.28	0.29	0.28	0.39
Na	0.3	0.14	0.17	0.29	0.22	0.1	0.14	0.39	0.41	0.4	0.15
Mg	0.25	0.31	0.35	0.21	0.29	0.38	0.33	0.27	0.32	0.38	0.37
K	1.4	1.48	2.35	1.97	1.98	1.36	1.45	2.91	1.56	1.51	2.28
Cl	0.89	0.71	0.95	0.85	0.92	0.47	0.44	1.56	0.97	0.61	1.04
S	0.27	0.22	0.47	0.22	0.29	0.35	0.26	0.32	0.42	0.31	0.25
DCAD[2]	72	107	116	258	167	40	143	280	45	37	202

[1]Calculated as 100 − CP − ether extract − NDF − ash.

[2]DCAD, mEq/kg of DM = [(Na% of DM/0.023) + (K% of DM/0.039)] − [(S% of DM/0.016) + (Cl% of DM/0.0355)].

Sample size calculations were performed using the G*Power software (version 3.1.9.2; University of Düsseldorf, Düsseldorf, Germany) to enable noninferiority analysis. The calculation regarding milk production was performed based on a previous study in which cows treated with cholecalciferol had lower milk production compared with untreated control cows (Venjakob et al., 2022). Assuming 80% power, a 95% confidence level, a SD of 8 kg, and a noninferiority limit of 2 kg, 198 animals per group were needed. For animal welfare reasons, a second sample size calculation was conducted to determine the number of animals needed to be sampled and detect a significant difference in serum total calcium (**tCa**) concentration. In the aforementioned study, cows treated with cholecalciferol had a 0.2 mmol/L higher serum tCa concentrations compared with untreated control cows (SD 0.3 mmol/L). Based on these results, a power of 80% and a 95% confidence level, it was required to take blood samples from 50 animals per group. For this purpose, a convenience sample of cows originating from 4 farms was selected based on the number of expected calvings and accessibility. Enrollment was not balanced with respect to the total number of animals included per farm, but efforts focused on enrolling approximately equal numbers of control and treatment cows within each farm. Only cows with a complete dataset (blood sample on 1, 2, and 3 DIM) were included in the statistical analyses, leading to exclusion of 7 cows. A formal sample size calculation was not conducted for the analyses of vitamin D metabolites. Instead, these analyses were performed in a convenience sample of cows, the size of which was determined a priori based on logistical feasibility and budgetary constraints, precluding expansion of this subset.

All herds participating in this study kept exclusively Holstein-Frisian cows and only multiparous cows were enrolled between April and October 2022. Inclusion criteria were (1) participation in a monthly DHIA equivalent testing system, and (2) feeding of a positive DCAD diet (>0 mEq/kg of DM) to prepartum cows. At the start of the study, the DCAD level was verified by a TMR sample analysis of each participating farm (Table 1). Average DCAD level was +133 mEq/kg (±81 SD; range 37 to 280 mEq/kg). Because 1 of the 12 farms started feeding a negative DCAD diet during the study period, it was excluded from the study. After enrollment, there was a 6-mo observation period (November 2022 to May 2023) to collect data on milk yield for each cow. The average herd size was 192 (±62 SD; range 60–280). Cows were milked in parallel (n = 8), or rotary (n = 1) milking parlors (twice daily). Two farms were using an automated milking system.

Dry cows were kept on deep straw (n = 2) or in freestall barns on cubicles equipped with straw and limestone and slatted floors (n = 9). Based on a random list generated in Excel (Microsoft Office Professional Plus 2016, Microsoft Deutschland Ltd.) cows were allocated to a treatment and a control group. Researchers were not blinded to the treatment allocation. Due to the previously observed adverse effects, cows of the treatment group were not treated with the recommended dose of 1×10^6 IU per 50 kg of BW but received a single injection with 10×10^6 IU of cholecalciferol intramuscularly (10 mL of Ursovit D_3, Serumwerk Bernburg, Bernburg, Germany). The injection was conducted on d 275 ± 2 of gestation to achieve an interval of 5 d between treatment and calving and cows were not reinjected. Cows of the control group received 10 mL of saline (NaCl 0.9%, B. Braun Melsungen AG, Melsungen, Germany) at the same time relative to expected calving to exclude negative effects caused by the injection itself. Data analyses at the start of the study showed that the average GL was 280 (±0.4 [SE]) d in the past 12 mo. As GL averaged 282 d during the study period, the interval between treatment and calving was 6.9 (±0.3) and 6.7 (±0.4) for cows in the treatment and the control group, respectively.

Blood Sampling and Analyses

A convenience sample of 95 cows from 4 different farms was blood sampled 0 to 24 h (d 1), 24 to 48 h (d 2), and 48 to 72 h (d 3) after calving. Additionally, in 30 out of these cows a blood sample was collected 2 d after cholecalciferol or placebo injection. Blood samples were drawn from the coccygeal vessels using sterile, evacuated serum collection tubes (8 mL, Vacuette, Greiner Bio-One GmbH, Frickenhausen, Germany) with a 20-gauge sterile cannula. Blood samples were allowed to clot for 30 min and then centrifuged at 21°C and 2,000 × *g* for 10 min to harvest serum (Hettich EBA 3S, Tuttlingen, Germany). Serum was separated into 3 aliquots and frozen at −18°C.

At the end of the study, the serum concentrations of tCa, P_i, and Mg were analyzed in the laboratory of the Farm Animal Clinic, Freie Universität Berlin. A Thermo Solaar M6 atomic absorption spectrometer (Thermo Fisher, Waltham, MA) was used to analyze the tCa concentration. The photometric analysis of the contents of Mg and P_i was carried out using a Cobas 8000 c701 analyzer (Roche Diagnostics International AG, Basel, Switzerland) and the test kits LT-Pi 0100 (P_i) and LTMG 0103 (Mg; Labor + Technik Eberhard Lehmann GmbH, Berlin, Germany). The interassay CV were 4.15% (tCa = 2.17 mmol/L; n = 30), 3.24% (P_i = 1.74 mmol/L; n = 30), and 7.51% (Mg = 0.88 mmol/L; n = 23). The intraassay coefficients of variation were 2.57% (tCa = 2.17 mmol/L; n = 30), 2.10% (P_i = 1.74 mmol/L; n = 30), and 2.85% (Mg = 0.88 mmol/L; n = 23).

Cholecalciferol, 25-OHD_3, and 24,25-dihydroxycholecalciferol (**24,25-$(OH)_2D_3$**) concentrations were determined at the Institute of Agricultural and Nutritional Sciences (Martin Luther University Halle-Wittenberg, Halle/Saale, Germany) by HPLC-MS/MS. A detailed description of the procedure has been published recently (Venjakob et al., 2024). The concentration of 1,25-$(OH)_2D_3$ was determined in a commercial laboratory (Immundiagnostik AG, Bensheim, Germany). The intra- and interassay CV were 6.7% and 9.0%, respectively, and the lower detection limit was 4.80 pg/mL.

At calving, calving ease (eutocia = calving without assistance, dystocia = calving assisted by 1 or more person), stillbirth (yes vs. no), and birth of twins (yes vs. no) were recorded. Within the first 5 DIM, cows were examined for RFM and clinical hypocalcemia. Cows were diagnosed with RFM when fetal membranes were not expelled by 24 h after calving. Cows were diagnosed with clinical hypocalcemia when they were recumbent before, at, or after calving and rose in response to an intravenous Ca infusion. Using the Metricheck device (Simcro, Hamilton, New Zealand), vaginal discharge was assessed once between 5 and 10 DIM. Clear or no mucus was categorized as score 0, cloudy mucus or mucus with flecks of pus as score 1, foul smelling and mucopurulent with ≤50% pus present as score 2, foul smelling and mucopurulent with >50% pus present as score 3, and putrid (foul smelling, watery, red or brown color) as score 4 (Urton et al., 2005). Cows with score 4 were diagnosed with metritis (Sheldon et al., 2006).

Statistical Analyses

Results from test-day data of the first 5 DHIA equivalent test days and blood analyses were combined (Access, Office 2010, Microsoft Deutschland Ltd., Munich, Germany), exported to Excel spreadsheets, and analyzed using SPSS for Windows (version 31.0, IBM Corp., Ehningen, Germany). To test whether parity, previous 305-d milk yield, GL, and interval between treatment and calving were evenly distributed among treatment and control cows, univariable models were calculated. Moreover, chi-squared tests were used to determine, whether treatment affected calving ease, the incidence of stillbirth, RFM, metritis, and clinical hypocalcemia.

Repeated measures ANOVA with first-order autoregressive covariance (GENLINMIXED procedure of SPSS) was performed to analyze the effect of treatment on milk production, postpartum blood minerals, and vitamin D metabolites. To meet model assumptions, concentrations of vitamin D metabolites were natural log-transformed before analysis, and least squares means were back-transformed and reported on the original scale. To evaluate the effect of the cholecalciferol treatment on milk yield, serum tCa, P_i, Mg, cholecalciferol, 25-OHD_3, 1,25-$(OH)_2D_3$, and 24,25-$(OH)_2D_3$ concentrations, 8 separate models were calculated. The individual cow was the experimental unit, and the farm was considered as a random effect. For the analysis of milk yield, repeated measures were conducted on test d 1 to 5. For serum tCa, P_i, and Mg concentrations repeated measures were conducted on 1, 2, and 3 DIM. For serum cholecalciferol, 25-OHD_3, 24,25-$(OH)_2D_3$, 1,25-$(OH)_2D_3$, repeated measures were conducted 2 d after cholecalciferol or placebo injection, and on 1, 2, and 3 DIM.

Each explanatory variable was separately analyzed in a univariable model (Dohoo et al., 2009). For the models of milk yield and blood minerals, the explanatory variables tested in the univariable models were treatment (cholecalciferol vs. saline), time (1, 2, 3 for blood minerals; test d 1, 2, 3, 4, 5 for milk production), parity (parity 2, parity 3, and parity ≥4), dystocia (yes vs. no), stillbirth (yes vs.

Table 2. Distribution of parity, milk yield in previous lactation, gestation length, and interval between treatment and calving, as well as incidence of eutocia, dystocia, stillbirth, retained fetal membranes, metritis, and clinical hypocalcemia (±SE) of cows treated once with 10 × 10^6 IU of cholecalciferol and cows treated with 10 mL of saline 7 d before calving considering all cows and those blood sampled

	All animals (n = 446)			Subsample of cows with blood samples (n = 95)		
Variable	Treatment group (n = 230)	Control group (n = 216)	*P*-value	Treatment group (n = 49)	Control group (n = 46)	*P*-value
Parity			0.58			0.80
2, n (%)	69 (30.0)	67 (31.0)		17 (34.7)	15 (32.6)	
3, n (%)	53 (23.0)	55 (25.5)		9 (18.4)	11 (23.9)	
≥4, n (%)	108 (47.0)	94 (43.5)		23 (46.9)	20 (43.5)	
Previous 305-d milk yield, kg	10,504 (±134)	10,260 (±133)	0.20	11,888 (±308)	11,226 (±282)	0.14
Gestation length, d	282.5 (±0.3)	282.4 (±0.4)	0.71	282.5 (±0.8)	282.2 (±0.7)	0.77
Δ treatment to calving, d	6.9 (±0.3)	6.7 (±0.4)	0.69	6.0 (±0.7)	6.7 (±0.9)	0.52
Calving ease[1]			0.30			0.75
Eutocia, n (%)	194 (84.3)	181 (83.8)		33 (67.4)	32 (69.6)	
Dystocia, n (%)	36 (15.7)	35 (16.2)		16 (32.6)	14 (30.4)	
Stillbirth, n (%)	9 (3.9)	3 (1.4)	0.10	2 (0.4)	1 (0.2)	0.60
Twins, n (%)	5 (2.2)	5 (2.3)	0.92	1 (2.0)	2 (4.3)	0.52
Retained fetal membranes,[2] n (%)	34 (14.8)	17 (7.9)	0.02	12 (24.5)	5 (10.9)	0.08
Metritis,[3] n (%)	7 (3.0)	3 (1.4)	0.22	5 (2.2)	3 (1.4)	0.05
Clinical hypocalcemia,[4] n (%)	29 (12.6)	21 (9.7)	0.33	4 (8.1)	3 (6.5)	0.74

[1]Eutocia = calving without any assistance; dystocia = calving assisted by 1 or more person.

[2]Cows were considered as having retained fetal membranes when fetal membranes were not expelled by 24 h after calving.

[3]Vaginal discharge score 4, according to the Metricheck examination.

[4]Cows were considered as having clinical hypocalcemia when they were recumbent before, at, or after calving and rose in response to an intravenous Ca infusion.

no), 305-d milk yield in previous lactation (continuous), as well as the interaction of time and treatment. For the vitamin D metabolites, the explanatory variables tested in the univariable models were treatment (cholecalciferol vs. saline), time (2 d after cholecalciferol or placebo injection and 1, 2, and 3 DIM), the interaction of time and treatment and parity (parity 2, parity 3, and parity ≥4).

If the univariable models resulted in a *P*-value <0.1, parameters were included in the mixed model. Using a backward stepwise elimination procedure that removed all variables with $P > 0.1$ from the model, selection of the model that best fit the data was performed. Interactions (time × treatment) were forced to remain in the model, whenever the inclusion led to a lower Akaike information criterion. To account for multiple comparisons, the *P*-value was adjusted using a Bonferroni correction in all models. Significance was declared when $P < 0.05$.

To evaluate the effect of treatment on probability of RFM and metritis a logistic regression model was built, using the GENLINMIXED procedure of SPSS. Cow was the experimental unit. Herd was considered as a random effect. Each explanatory variable was separately analyzed in an univariable model. Except for time, we included the same variables as fixed effects, as for the models described above. Additionally, further calving related variables, such as birth of twins (yes vs. no) and sex of the calf (female vs. male) was considered in this model. Selection of the model that best fit the data was performed by using a backward stepwise elimination procedure that removed all variables with $P > 0.10$ from the model.

RESULTS

Parity, previous 305-d milk yield, GL, and interval from treatment to calving were evenly distributed between control and treatment group (Table 2). Furthermore, calving ease, incidence of clinical hypocalcemia and metritis were not affected by the treatment.

Milk Production

The average 305-d milk production on the farms included was 10,614 (±694 SD; range 9,278–11,700). Figure 1 shows the ECM yield of the treatment and control group over the first 5 test days after calving. Milk production was associated with test day ($P < 0.01$), parity ($P < 0.01$), previous 305-d milk yield ($P < 0.01$), and stillbirth ($P < 0.05$). Treatment ($P = 0.53$) and the interaction of treatment and time of test day ($P = 0.73$) had no effect on milk yield. Cows treated with cholecalciferol had an ECM production of 38.8 (95% CI 37.2 to 40.6) kg, 39.5 (95% CI 37.8 to 41.2) kg, 38.1 (95% CI 36.3 to 39.8) kg, 37.1 (95% CI 35.4 to 38.8) kg, and 35.8 (95% CI 34.1 to 37.5) kg at first, second, third, fourth, and fifth test day, respectively. Energy-corrected milk production of the control group was 39.1 (95% CI 37.4 to 40.9) kg, 39.8 (95% CI 38.1 to 41.6) kg, 38.4 (95% CI 36.7 to

Table 3. Final multivariable logistic regression model evaluating the effect of prepartum injection with cholecalciferol on the probability of retained fetal membranes[1]

Variable	Estimate	SE	95% CI Lower CI	95% CI Upper CI	Odds ratio	*P*-value
Intercept	−2.71	0.31	−3.31	−2.10	0.07	<0.01
Treatment						
Control[2]	Referent					
Treatment[3]	0.75	0.34	0.09	1.41	2.12	<0.05
Stillbirth						
Alive	Referent					
Dead	1.37	0.74	−0.09	2.82	3.91	0.07
Twins						
Singleton	Referent					
Twins	2.76	0.79	1.21	4.31	15.96	<0.01

[1]Cows were diagnosed with RFM when fetal membranes were not expelled by 24 h after calving.

[2]Cows of the control group were injected with 10 mL of saline 7 d before calving.

[3]Cows of the treatment group were injected with 10×10^6 IU cholecalciferol 7 d before calving.

40.2) kg, 37.0 (95% CI 35.2 to 38.8) kg, and 36.3 (95% CI 34.5 to 38.0) kg at first, second, third, fourth, and fifth test day, respectively.

Uterine Health

Cows treated with 10×10^6 IU of cholecalciferol had a greater incidence of stillbirth compared with control cows (3.9% vs. 1.4%; $P = 0.10$). Cows of the treatment group had a greater incidence of RFM (14.8% vs. 7.9%; $P < 0.05$; Table 2). To evaluate the risk of treated cows to develop RFM and metritis a logistic regression model was calculated. Although twin calving had the greatest effect on the risk of RFM (odds ratio: 15.86; $P < 0.01$; 95% CI 3.37 to 74.72; Table 3), the prepartum treatment with cholecalciferol (odds ratio: 2.12; 95% CI 1.10 to 4.09; $P < 0.05$) and the occurrence of stillbirth (odds ratio: 3.92; 95% CI 0.92 to 16.75; $P = 0.07$) also effected the risk of RFM (Table 3). In contrast, prepartum treatment with cholecalciferol had no effect on the risk of developing metritis.

Blood Minerals

In both groups, serum tCa concentration was lowest on 1 DIM and then increased steadily until 3 DIM to 2.1 mmol/L (Figure 2A). Accordingly, the time of sampling relative to calving was associated with serum tCa concentration ($P < 0.01$). Cows injected intramuscularly with 10×10^6 IU cholecalciferol 7 d before the expected calving date had higher serum tCa concentrations over the first 3 DIM (2.00 [95% CI 1.95 to 2.05] mmol/L) compared with animals treated with 10 mL of saline (1.93 [95% CI 1.88 to 1.98] mmol/L; $P = 0.08$). The maximum difference observed was 0.1 mmol/L at 1 DIM. As expected, parity was significantly associated with serum tCa concentrations ($P < 0.01$) with lower serum tCa concentrations in older cows.

The serum P_i concentration also increased from 1 to 3 DIM with the nadir occurring at 1 and 2 DIM in cholecalciferol treated and control cows, respectively (Figure 2B). Accordingly, the time of sampling was significantly associated with serum P_i concentration ($P < 0.05$). At all 3 sampling times, the animals in the treatment group had significantly higher serum P_i concentrations than the animals in the control group ($P < 0.05$). Although cows treated with cholecalciferol had P_i concentrations of 1.45 (95% CI 1.31 to 1.59) mmol/L, 1.51 (95% CI 1.38 to 1.64) mmol/L, and 1.60 (95% CI 1.48 to 1.74) mmol/L on 1, 2, and 3 DIM, respectively, the concentration of saline-treated cows were of 1.32 (95% CI 1.18 to 1.46) mmol/L, 1.31 (95% CI 1.18 to 1.44) mmol/L, and 1.46 (95% CI 1.32 to 1.59) mmol/L on 1, 2, and 3 DIM, respectively. Furthermore, there was an association between serum P_i concentration and parity ($P < 0.01$), with older cows having lower P_i concentrations.

Magnesium concentrations over the first 3 DIM showed an inverse pattern compared with those of P_i and tCa (Figure 2C). Cows treated with cholecalciferol had lower serum Mg concentrations over the first 3 DIM compared with cows of the control group ($P < 0.01$). In cows treated with cholecalciferol serum Mg concentrations on 1, 2, and 3 DIM were 0.89 (95% CI 0.84 to 0.93) mmol/L, 0.89 (95% CI 0.85 to 0.93) mmol/L and 0.82 (95% CI 0.78 to 0.86) mmol/L, respectively. Cows of the control group had serum Mg concentrations on 1, 2, and 3 DIM of 1.03 (95% CI 0.98 to 1.07) mmol/L, 0.99 (95% CI 0.95 to 1.04) mmol/L and 0.90 (95% CI 0.86 to 0.94) mmol/L, respectively. The lowest concentration was measured on 3 DIM. Accordingly, the time of sampling relative to calving was associated with serum Mg concentration ($P < 0.01$).

Vitamin D Metabolites

In a subsample of 30 cows (15 treated with cholecalciferol, 15 treated with saline, 7 d before calving) blood samples were obtained 2 d after application of cholecalciferol or saline and on 1, 2, and 3 DIM. In those cows, concentrations of cholecalciferol, 25-OHD$_3$, 1,25-(OH)$_2$D$_3$, and 24,25-(OH)$_2$D$_3$ were determined (Figure 3).

Two days after cholecalciferol injection, the average serum cholecalciferol concentration in the control group was 3.1 (95% CI 1.9 to 4.9) ng/mL. In the treatment group, average cholecalciferol concentration was approximately 150 times higher at 466.6 (95% CI 295.3 to 737.2) ng/mL. After calving, the concentration decreased to 80.7 (95% CI 51.1 to 127.5), 64.1 (95% CI 40.6 to 101.3), and 48.7 (95% CI 30.8 to 77.0) ng/mL on d 1, 2, and 3, respectively. The cholecalciferol concentration of the control group remained low after calving (2.3 [95% CI 1.4 to 3.6], 2.5 [95% CI 1.6 to 4.0], and 2.3 [95% CI 1.5 to 3.7] ng/mL on d 1, 2, and 3, respectively; Figure 3A).

The concentration of 25-OHD$_3$ was approximately 3 times higher in the treatment group, compared with control group 2 d after the cholecalciferol injection (148.3 [95% CI 127.1 to 173.1] and 48.1 [95% CI 41.1 to 56.3] ng/mL for treated and control cows, respectively; $P <$ 0.01). Although the concentration remained at 49.5 (95% CI 42.3 to 57.9), 49.3 (95% CI 42.1 to 57.7), and 48.2 (95% CI 41.2 to 56.4) ng/mL on 1, 2, and 3 DIM in control cows, respectively, 25-OHD$_3$ concentration increased to 194.2 (95% CI 166.4 to 226.6), 203.2 (95% CI 174.1 to 237.1), and 195.2 (95% CI 167.3 to 227.8) ng/mL on 1, 2, and 3 DIM in treated cows, respectively, (Figure 3B).

Two days after the cholecalciferol or placebo injection, cholecalciferol-treated cows had increased 1,25-(OH)$_2$D$_3$ concentrations (67.1 [95% CI 48.3 to 93.3] pg/mL) compared with control cows (32.4 [95% CI 24.4 to 43.16] pg/mL; Figure 3C). After calving, the concentration in the control group tripled (92.2 [95% CI 69.3 to 122.7] pg/mL), whereas the concentration in the treatment group decreased slightly in comparison to the prepartum concentration (60.1 [95% CI 43.2 to 83.5] pg/mL). In the control group, the 1,25-(OH)$_2$D$_3$ concentration peaked on 2 DIM (134.2 [95% CI 100.9 to 178.6] pg/mL) and remained high on 3 DIM (134.0 [95% CI 100.7 to 178.3] pg/mL). In the treatment group, the 1,25-(OH)$_2$D$_3$ concentration increased slightly on 2 DIM (65.06 [95% CI 46.8 to 90.48] pg/mL) and 3 DIM (74.1 [95% CI 53.3 to 103.1] pg/mL).

Two days after treatment application, the 24,25-(OH)$_2$D$_3$ concentration was twice as high in the treatment group (6.0 [95% CI 4.5 to 7.8] ng/mL) compared with control animals (2.9 [95% CI 2.2 to 3.8] ng/mL). Over the first 3 DIM, the 24,25-(OH)$_2$D$_3$ concentration remained this low in the control group (2.8 [95% CI 2.1 to 3.7], 2.9 [95% CI 2.2 to 3.7], and 2.8 [95% CI 2.2 to 3.7] ng/mL on 1, 2, 3 DIM, respectively), whereas a further increase to values >20 ng/mL was observed in treated cows (19.9 [95% CI 15.2 to 26.0], 22.1 [95% CI 16.9 to 29.0], and 23.2 [95% CI 17.7 to 30.3] ng/mL on 1, 2, and 3 DIM, respectively; Figure 3D).

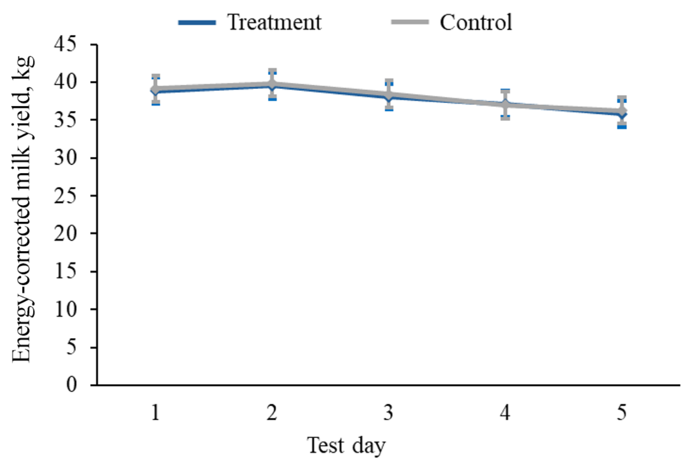

Figure 1. Effect of treatment with 10 × 10^6 IU of cholecalciferol (n = 230) or placebo (n = 216; 10 mL of saline) 7 d before expected parturition on milk production over the first 5 test days using least square estimates (mean, 95% CI) from repeated-measure ANOVA. Number of test day (P < 0.01), parity (P < 0.01), 305-d milk yield of previous lactation (P < 0.01), and stillbirth (P < 0.05) were significantly associated with milk production during the first 5 test day. Treatment (P = 0.53) and number of test day by treatment (P = 0.73) did not affect milk production.

DISCUSSION

The objective of the present study was to evaluate the effect of a single prepartum injection of 10 × 10^6 IU cholecalciferol on postpartum milk production, uterine health, blood mineral concentrations and vitamin D metabolites on commercial dairy farms feeding a positive prefresh DCAD diet. No difference was observed in milk yield over the first 5 test days but cows treated with cholecalciferol had a higher risk for RFM. There were no differences between the treatment and control group with regard to calving ease and the incidence of clinical hypocalcemia. Evaluating serum tCa concentration over the first 3 DIM, a minor positive effect was observed. Analysis of the concentrations of cholecalciferol, 25-OHD$_3$, and 24,25-(OH)$_2$D$_3$ 2 d after cholecalciferol injection and on 1, 2, and 3 DIM, showed a significant increase in vitamin D metabolites already on d 2 after treatment. Also, 1,25-(OH)$_2$D$_3$ concentration was affected by treatment with concentrations being increased 2 d after injection and decreased on 1, 2, and 3 DIM.

Milk Production

The evaluation of milk yield over the first 5 test days showed no effect of intramuscular cholecalciferol

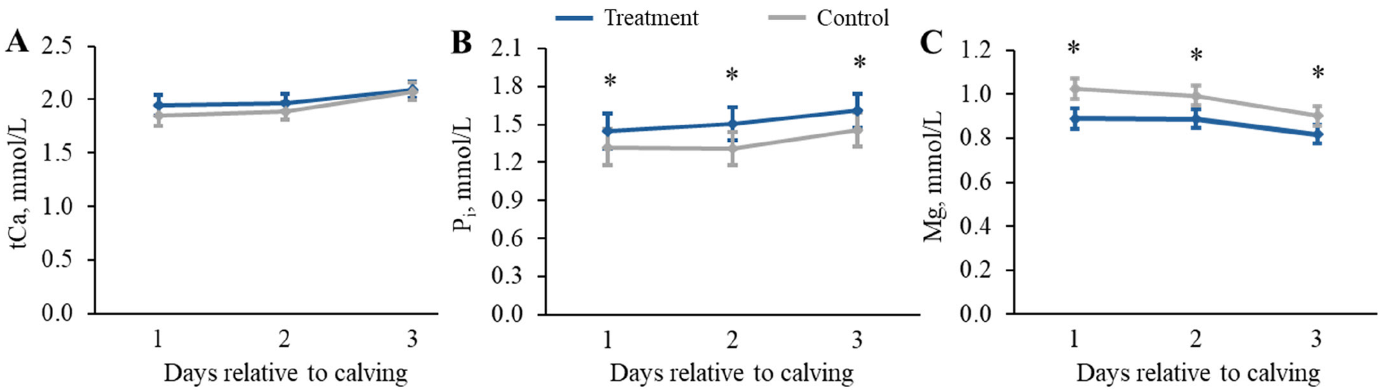

Figure 2. Effect of treatment with 10×10^6 IU of cholecalciferol (n = 49) or placebo (n = 46; 10 mL of saline) 7 d before expected parturition on serum total calcium (tCa), phosphate (P_i), and Mg concentration over the first 3 DIM using least square estimates (mean, 95% CI) from repeated-measure ANOVA. The final model for serum tCa concentration included time ($P < 0.01$), treatment ($P = 0.08$), parity ($P < 0.01$), and time by treatment ($P = 0.57$). The final model for serum P_i concentration included time ($P < 0.05$), treatment ($P < 0.05$), parity ($P < 0.05$), and time by treatment ($P = 0.85$). The final model for serum Mg concentration included time ($P < 0.01$), treatment ($P < 0.01$), and time by treatment ($P = 0.50$). Significant differences are indicated by an asterisk at each time point.

treatment before calving. To our knowledge, no other group of authors has investigated the effect of parenteral cholecalciferol treatment on milk production. The decrease of milk yield associated with cholecalciferol treatment in previous studies (Venjakob et al., 2022, 2024) was probably related to adverse effects on health status and could not be reproduced in this multisite setting. Studies exist that evaluated milk production of cows treated with cholecalciferol and 25-OHD_3, orally. Martinez et al. (2018b) investigated the effect of different DCAD levels (+130 or −130 mEq/kg DM) and oral application of 2 vitamin D metabolites (3 mg cholecalciferol or 25-OHD_3 per 11 kg of DM) on lactation performance. In this study feeding of 25-OHD_3 had a positive effect on milk production until 49 DIM. This finding was replicated by Silva et al. (2022) who also observed increased milk production in cows fed with a prepartum negative DCAD diet in combination with 3 mg of 25-OHD_3. In another study, evaluating the effect of feeding 6 mg of 25-OHD_3 daily in combination with a negative DCAD diet during the last 13 d of gestation, no effect on milk yield was observed during the first 28 d of lactation (Weiss et al., 2015).

Stillbirth and RFM

Cholecalciferol treatment had a negative effect on uterine health. In treated cows, we observed a significantly greater incidence of RFM and a trend for a greater incidence of stillbirth. Also, the incidence of metritis was increased in the treatment group (3.0 vs. 1.4%). Cows treated with 10×10^6 IU cholecalciferol had a 2.12 greater risk of incurring RFM. Martinez et al. (2018a) also evaluated the effect of a combined feeding of a positive or negative DCAB (± 130 mEq/kg DM) with oral supplementation of 25-OHD_3 or cholecalciferol on health and reproduction. In their study, cows fed with cholecalciferol showed a significantly greater incidence of RFM and metritis. This is in agreement with our previous studies in which treatment with cholecalciferol negatively affected the risk for RFM, metritis (Venjakob et al., 2022, 2024), and time to pregnancy (Venjakob et al., 2022). Based on these findings, we suspected that these effects might have been evoked by the shortened GL observed in treated cows. As the control cows had been left untreated, interference with the stress of the prepartum injection could not be excluded. In the present study, we therefore implemented a placebo treatment as control and did not observe differences in GL in comparison of treatment and control group. The incidence of RFM, however, still differed.

The separation of the maternal and fetal tissues of the placentome after calving is a complex process involving the activation of a local immune response (Attupuram et al., 2016). A comparison of the mRNA expression profile in placental tissues obtained from cows that developed RFM and healthy controls revealed a decreased expression of pro-inflammatory cytokines such as IL-1, IL-6, IL-8, and TNF-α (Boro et al., 2014). In human placental cells, IL-6 and TNF-α were downregulated by 1,25-$(OH)_2D_3$ (Noyola-Martínez et al., 2013). Interestingly, a modulation of the expression of pro-inflammatory cytokines was also seen after treatment with cholecalciferol in the placenta of mice challenged with LPS (Chen et al., 2015). To investigate whether cholecalciferol and its metabolites interfere with the separation and thus the expulsion of the fetal membranes, further studies are needed.

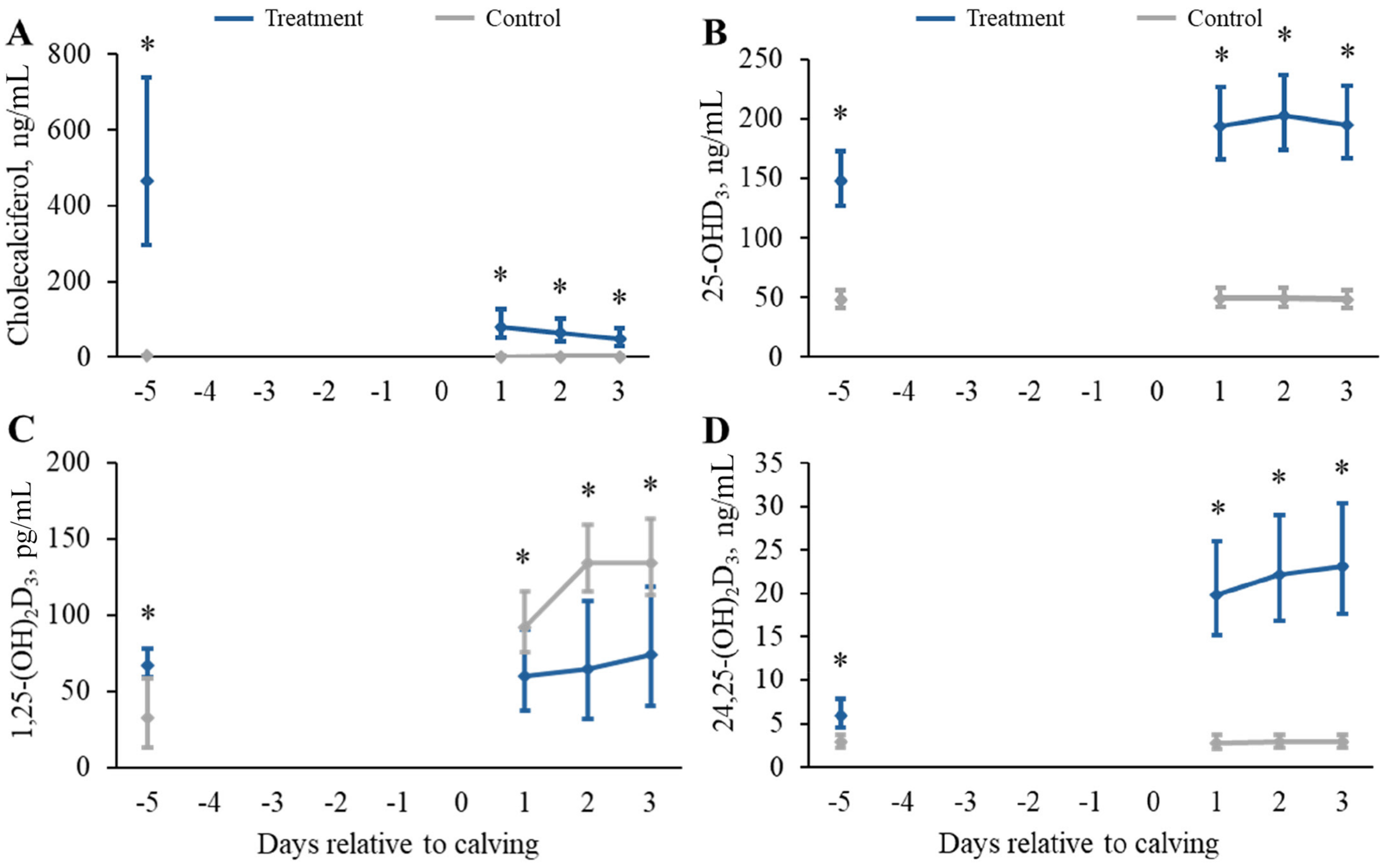

Figure 3. Effect of treatment with 10×10^6 IU of cholecalciferol (n = 15) or placebo (n = 15; 10 mL of saline) 7 d before expected parturition on cholecalciferol, 25-hydroxycholecalciferol (25-OHD_3), 1,25-dihydroxycholecalciferol (1,25-$(OH)_2D_3$), and 24,25-dihydroxycholecalciferol (24,25-$(OH)_2D_3$) over the first 3 DIM using least square estimates (mean, 95% CI) from repeated-measure ANOVA. For the analyses, data were natural log-transformed and LSM were subsequently back-transformed. In the models of cholecalciferol, 25-OHD_3, and 24,25-$(OH)_2D_3$ time ($P < 0.01$), treatment ($P < 0.01$), and the interaction of time by treatment ($P < 0.01$) remained in the final models. In the model of 1,25-$(OH)_2D_3$ time ($P < 0.01$), treatment ($P < 0.05$), parity ($P < 0.05$), and the interaction of time by treatment ($P < 0.01$) remained in the final model. Significant differences are indicated by an asterisk at each time point.

Blood Minerals

In contrast to our previous studies, the mean serum tCa concentration was below 2.0 mmol/L over the first 48 h after calving, independent from prepartum treatment (cholecalciferol or saline). On 3 DIM, the mean tCa concentration in both groups was at a similar level of approximately 2.1 mmol/L. This observation is in line with previous findings (Kimura et al., 2006; Megahed et al., 2018) as activation of Ca metabolism takes up to 48 h to be effective (Martín-Tereso and Martens, 2014). In addition, feed intake is decreased around calving, leading to reduced Ca uptake via the diet (Huzzey et al., 2007; Neave et al., 2017). Cows in the treatment group showed only numerically higher tCa concentrations after calving. The maximum difference was observed on 1 DIM where cholecalciferol-treated cows had 0.1 mmol/L higher concentrations of tCa. As only Ca^{2+} is available to the cow and the ratio of tCa and Ca^{2+} is approximately 2:1 (Cohrs et al., 2023) the biological relevance of this finding is questionable. Data from literature is inconsistent. In a study from Japan (Naito et al., 1987), lactating Holstein-Frisian cows were injected on d 7 after calving with 10×10^6 IU of cholecalciferol. Plasma concentration of tCa increased and was significantly higher than in untreated control cows from d 2 to 15 after treatment. Zepperitz et al. (1991) injected nonlactating cows with 10×10^6 IU of cholecalciferol, 4 mg of 25-OHD_3, or 1-α-hydroxycholecalciferol, 3 times in a weekly interval and observed highest tCa concentrations in the cholecalciferol group reaching concentrations above 3.0 mmol/L already after the first cholecalciferol treatment. Because these results were obtained in nonpregnant cows, they are difficult to compare with the present study, as the transition from late pregnancy to lactation has a significant effect on Ca and vitamin D metabolism. However, there is a recent study that reported decreased concentrations of Ca^{2+}, PTH, and bone markers following a treatment with 8×10^6 IU of cholecalciferol, intramuscularly. These cows were fed with a negative DCAD close-up diet (−81 mEq/kg DM) and treated 2 to 8 d prepartum in comparison to a control group (Sadri et al., 2021). This outcome is surprising as feeding of a negative DCAD diet results in metabolic acidosis that increases PTH secretion (Lopez et al., 2002) and increases PTH sensitivity of tissues (Goff et al., 2014) probably leading to an enhanced conversion of 25-OHD$_3$ to 1,25-(OH)$_2$D$_3$. We speculated that this mechanism explained the stronger effect on peripartum Ca homeostasis observed in our earlier studies using a negative DCAD diet (Venjakob et al., 2022, 2024). In those studies, tCa concentrations in control cows were relatively low (i.e., 1.63 mmol/L and 1.71 mmol/L) compared with 1.85 mmol/L at 1 DIM in the current study. Cows treated with cholecalciferol (12 × 10^6 IU, 6 × 10^6 IU, 12 × 10^6 IU, and 10 × 10^6 IU) had tCa concentrations of 1.95, 1.81, 2.05, and 1.95 mmol/L, respectively. Because control cows in the present study had higher baseline tCa, the effect of cholecalciferol injection was modest, with a maximum difference of 0.1 mmol/L at 1 DIM. Studies evaluating the effect of feeding 25-OHD$_3$ were also able to increase the 25-OHD$_3$ concentrations to 200 ng/mL (Wilkens et al., 2012; Rodney et al., 2018; Silva et al., 2022). The effect of feeding 25-OHD$_3$ on postpartum Ca concentration, however, were conflicting. Although some authors found effects on Ca^{2+} (Wilkens et al., 2012; Silva et al., 2022), others found no effect (Rodney et al., 2018). Interestingly, 1,25-(OH)$_2$D$_3$ was not increased after oral 25-OHD$_3$ supplementation in most of the studies. In contrast, it has been shown that the parenteral application of 300 µg of 1,25-(OH)$_2$D$_3$ substantially increased Ca^{2+} and tCa concentrations (Vieira-Neto et al., 2021). This effect seemed to be mediated, however, by supraphysiological concentrations of 1,25-(OH)$_2$D$_3$ of >450 pg/mL. In this study, tCa concentrations remained increased until 7 DIM compared with control cows. Between 9 and 15 DIM, however, the concentration of tCa was lower than in control cows.

Cows treated with cholecalciferol had higher serum P_i concentrations and lower Mg concentrations compared with control cows. This finding confirms results of previous studies (Venjakob et al., 2022, 2024; Poindexter et al., 2023b). Renal excretion of Mg depends on oral Mg intake and the concentration of PTH (Martens et al., 2018). As PTH concentrations might have been higher in the control group, these findings may support the assumption, that PTH secretion raises the threshold for renal Mg excretion, resulting in a higher serum Mg concentration (Goff, 2008).

Vitamin D Metabolites

Cows treated with 10 × 10^6 IU of cholecalciferol showed 150-fold increased serum cholecalciferol concentrations 2 d after intramuscular injection compared with cows in the control group. After calving, the concentration dropped considerably, but was still 30-fold increased, compared with the control group (Figure 3A). A comparable increase was also observed in a previous study; however, in that study the cholecalciferol concentration of cows treated with 12 × 10^6 IU of cholecalciferol was above 400 ng/mL on the day after calving (Venjakob et al., 2024). One reason for the higher cholecalciferol concentration may be the higher dosage of 12 × 10^6 IU of cholecalciferol. Conversely, the interval between treatment and calving was 2 d shorter, which may also explain the lower cholecalciferol concentrations in the present study. To the authors' knowledge, no further studies exist that evaluated serum cholecalciferol concentrations following cholecalciferol injection. Other authors investigated plasma cholecalciferol concentrations after oral supplementation of high amounts of cholecalciferol (Rodney et al., 2018). In this study, cows fed with 120,000 IU (DCAD + 130 mEq/kg) cholecalciferol starting at d 255 of gestation had the highest cholecalciferol concentrations of approximately 17 ng/mL during the last days before calving.

The concentration of 25-OHD$_3$ was also considerably increased in the treatment group, compared with the control group (183.9 vs. 48.8 ng/mL). Although the concentration remained low in control cows, the 25-OHD$_3$ concentration after calving stabilized at approximately 200 ng/mL in treated cows (Figure 3B). These results are comparable to the ones provided by Naito et al. (1987), who injected cows intramuscularly with 10 × 10^6 IU cholecalciferol on 7 DIM. Two days after the treatment, the authors determined higher 25-OHD$_3$ concentrations of approximately 200 ng/mL and concentrations around 300 ng/mL 6 to 8 d following treatment. Conversion of cholecalciferol to 25-OHD$_3$ in the liver is attributed to microsomal CYP2R1 and mitochondrial CYP27A1 and is not regulated tightly, meaning that it is mainly dependent on the concentration of the substrate (Jones, 2008). Littledike and Horst (1982) treated Jersey cows with 15 × 10^6 IU of cholecalciferol 1 mo before calving and 2.5 × 10^6 IU 1 wk later. Those authors observed similar 25-OHD$_3$ concentrations to ours but also hypercalcemia with concentrations up to 3.0 mmol/L tCa. Consequently, clinical signs of vitamin D toxicity, such as inappetence, polypnea with pronounced expiration, stiffness of joints, and recumbency were observed and 10 out of 17 treated animals died. In contrast, Jones (2008) considered plasma 25-OHD$_3$ concentration as a good biomarker for vitamin D toxicity, however claiming around 300 ng/mL as a potential threshold. None of the cows included in the present study showed hypercalcemia or classical symptoms of vitamin D toxicity.

In the kidney, 25-OHD$_3$ is hydroxylated to 1,25-(OH)$_2$D$_3$. This process, attributed to CYP27B1 mainly depends on blood concentrations of Ca^{2+} and PTH. However, 2 d after cholecalciferol injection, the concentration of 1,25-$(OH)_2D_3$ was increased compared with cows of the control group in the present study (Figure 3C). At first sight, this finding is surprising, as there was no stimulus to activate CYP27B1 before calving. However, Naito et al. (1987) observed a transient increase in 1,25-$(OH)_2D_3$ concentrations for 3 d after an intramuscular administration of cholecalciferol to lactating cows. Rodney et al. (2018) observed the same effect in prepartum cows fed 3 mg of 25-OHD_3 and assumed that supplementation with large amounts of 25-OHD_3 may override regulatory mechanisms for 1,25-$(OH)_2D_3$ synthesis. One might speculate that the organism's regulatory circuits may transiently be overcome by an increased production of 1,25-$(OH)_2D_3$ via extrarenal CYP27B1. Kägi et al. (2018) showed in mice that an injection of 1,25-$(OH)_2D_3$ did only affect renal expression but not extrarenal expression of CYP27B1.

In line with our previous study (Venjakob et al., 2024), plasma concentrations of 1,25-$(OH)_2D_3$ were decreased in the treatment group, although tCa was slightly increased. We assume the postpartum concentrations of tCa were maintained by the high plasma concentrations of 25-OHD_3. The binding affinity of 25-OHD_3 to the vitamin D binding protein (**VDBP**) is higher compared with 1,25-$(OH)_2D_3$ (Jones et al., 2006). As only unbound 1,25-$(OH)_2D_3$ can diffuse into the cells to bind the intracellular VDR, an effect on target tissues can probably be exerted by elevated concentrations of free 1,25-$(OH)_2D_3$ due to a displacement from the VDBP by 25-OHD_3 (Jones, 2008; Lütke-Dörhoff et al., 2022). In addition, 25-OHD_3 itself can bind to the VDR if present at concentrations >150 ng/mL (DeLuca et al., 2011; Quesada-Gomez and Bouillon, 2018).

Both metabolites, 25-OHD_3 and 1,25-$(OH)_2D_3$ are inactivated by hydroxylation at position 24 attributed to CYP24A1 (Jones et al., 2012). This enzyme is activated by 1,25-$(OH)_2D_3$, FGF23, Ca^{2+} and P_i, and inhibited by PTH (Dusso et al., 2005; Christakos et al., 2019). The induction of CYP24A1 by the treatment is indicated by the higher concentrations of 24,25-$(OH)_2D_3$ observed already 2 d after the administration. After calving, concentrations further increased in cows treated with cholecalciferol. A stimulated expression of renal CYP24A1 accompanied by a downregulation of the expression of CYP27B1 in response to treatment with different vitamin D metabolites has been shown before in several species including sheep and goats (Herm et al., 2015; Wilkens et al., 2016).

Study Limitations

The sample size calculation conducted assumed independence among cows because reliable intraclass correlation coefficient estimates were unavailable at the time of study planning. Post hoc analysis, however, revealed a substantial within farm correlation. Consequently, the statistical power is limited and the risk to incorrectly rule out small differences in milk yield between groups is increased.

We aimed to treat cows enrolled in this study approximately 5 d before expected parturition. Due to an increased GL during the study period, the interval between treatment and calving was 7 d. The manufacturer of the cholecalciferol product used recommends reinjecting cows with the same amount of cholecalciferol (1×10^6 IU per 50 kg of BW) if the cow does not calve within 6 d after first treatment and some authors speculate that there may be a negative effect on Ca homeostasis when the treatment exceeds this interval (Littledike and Horst, 1980). The present data show, however, that even after an interval of 7 d, the concentration of 25-OHD_3 was still 3 times higher than in the control group, indicating that the treatment was sufficiently close to calving.

In comparison to previous studies, we included a placebo injection of control cows with 10 mL saline solution. Ideally, placebo injections should be conducted with the identical product except for the active substance (cholecalciferol) to exclude effects caused by the adjuvants. Before the start of the study period, we explored the production of such a placebo. However, due to pharmaceutical regulations and production-related issues this approach was not implemented.

Blood samples were drawn from a subsample of 95 cows of 4 different farms. Those farms were selected as it was expected that several cows would calve within a short period of time, and as these farms were easily accessible. Unfortunately, for reasons of time and cost, no blood sample was taken before treatment; therefore, there is no data on baseline concentrations of minerals and vitamin D metabolites.

CONCLUSIONS

The present study shows minor effects of a prepartum cholecalciferol injection on blood minerals in cows that were fed a positive DCAD in the prefresh diet. Although milk production was not affected by the treatment, we observed a greater incidence of RFM in cows treated with cholecalciferol. Based on these results on farms feeding a positive DCAD diet, it is questionable whether the minor positive effects on Ca concentration, accompanied by adverse effects on uterine health, justify the effort involved in this prophylactic treatment. Prepartum cholecalciferol injection increased 1,25-$(OH)_2D_3$ 2 d after injection, presumably by overstraining of regulatory mechanisms due to the high dosage. On 2 and 3 DIM the concentration of 1,25-$(OH)_2D_3$ was, however, higher in the control group. Based on the results of vitamin D metabolites we suspect that the postpartum increase in tCa in the treatment group is mediated by a sustained increase of 25-OHD$_3$ concentration.

NOTES

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Nonstandard abbreviations used: 25-OHD$_3$ = 25-hydroxycholecalciferol; 1,25-(OH)$_2$D$_3$ = 1,25-dihydroxycholecalciferol; 24,25-(OH)$_2$D$_3$ = 24,25-dihydroxycholecalciferol; Ca^{2+} = ionized Ca; FGF23 = fibroblast growth factor 23; GL = gestation length; P$_i$ = inorganic phosphate; PTH = parathyroid hormone; RFM = retained fetal membranes; tCa = total Ca; VDBP = vitamin D binding protein; VDR = vitamin D receptor.

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